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Lynda Lukasik, Director of Climate Change Initiatives

Office of Climate Change Initiatives, City of Hamilton

71 Main Street West, Hamilton, Ontario L8P 4Y5

Dear Lynda Lukasik,

My name is Marchelino Toma, and I am a student in SUSTAIN 2SS3, Advocating for Sustainability, at McMaster University. This term, I am working with four classmates, Nicholas Gyorgypal, Karanveer Sandhu, Saha Rafiei, and Mehraban Shahbazi, to support the City of Hamilton Forestry Section with “If Trees Could Talk,” the tree monitoring pilot it runs in partnership with your Office (City of Hamilton, n.d.). By December 4, 2026, we will give Forestry staff a research report, an advocacy plan, and at least 20 social media posts in the trees’ voices. Our main focus is data stewardship, meaning how the pilot’s readings are collected, stored, and shared. My research looks at how heat, drought, and storms affect urban forests.

I am writing because this pilot has already shown that residents will listen when a tree has something to say, yet the future of the data behind those messages is still uncertain. If the sensors go quiet when the pilot period ends, Hamilton will lose a local record of how its trees cope with heat and dry spells just as that record becomes useful. I am therefore asking your Office to support continued Climate Change Reserve funding for the pilot in the Reserve’s next intake and to make sharing its data on Open Hamilton, the City’s open data portal, part of that funding.

Since last fall, sensors on a silver maple in Victoria Park and a black walnut in Captain Cornelius Park have tracked soil moisture at the roots, how much water each tree takes up, and

the swelling and shrinking of the bark (City of Hamilton, n.d.). Residents chose their names in a public contest, so Hamilton now knows them as Acer the Carbon Eraser and Bark Vader. The City turns the readings into posts in each tree's voice, and at our first meeting on October 1, Forestry staff suggested we write more. For residents, hearing from a thirsty maple in a familiar park makes a heat wave feel more real than any chart. Because the Climate Change Reserve helped fund the pilot, it already sits within your Office's climate work (City of Hamilton, n.d.).

This matters because Hamilton's trees face growing climate pressure. A study of 3,129 urban tree and shrub species across 164 cities found that 56% already experience temperatures beyond those found in their natural range, and that share is projected to reach 76% by 2050 under an intermediate emissions scenario (Esperon-Rodriguez et al., 2022). Locally, canopy cover inside the urban boundary has stayed at 18% since 2021, less than half of the 40% goal for 2050, and central and east wards sit below the city average (City of Hamilton, 2026). Residents there have less shade on the hottest days. To close that gap, Forestry staff must decide which species can cope with more heat and drought and where to plant them. The two pilot trees allow a local comparison between an urban park and a natural site. However, one year of readings cannot show whether a dry July was unusual or the start of a pattern. That takes several seasons of data.

For now, the public mainly sees the readings through the trees' posts and a dashboard. An early 2024 version of this idea, developed through CityLAB Hamilton, planned to make all collected data available through the City's open data program (CityLAB Hamilton, 2024). This year, the Forestry Section asked CityLAB how the pilot's data could support research and academic work (CityLAB Hamilton, 2026). Harvard Forest offers a useful model. Its talking red oak draws on more than 55 years of archived climate records, so each post rests on decades of evidence (Harvard Forest, n.d.). Hamilton can do the same with staff time rather than new

equipment, by posting the readings on Open Hamilton every three months with a short guide to each sensor and any gaps, while CityLAB keeps matching students to the data. Later, the same rule could apply to every Reserve project that collects environmental data. Residents would keep a voice they helped name, and researchers and teachers would gain trusted local records.

In short, the data behind these trees should outlast the pilot, so I am asking your Office to take three steps. First, please support renewed Climate Change Reserve funding in the next intake so both trees keep reporting through 2027. Second, please make quarterly publication on Open Hamilton, with a short data guide, a condition of that funding. Third, please reply in writing by Friday, November 13, 2026, or meet with our group for 20 minutes on Microsoft Teams before November 27, so our final recommendations to the Forestry Section on December 4 reflect what your review team looks for. Thank you for considering this request.

Sincerely,

Marchelino Toma

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cc Katie Mayne, Senior Project Manager, Forestry and Horticulture, City of Hamilton

References

City of Hamilton. (n.d.). *If Trees Could Talk*. Engage Hamilton. https://engage.hamilton.ca/IfTreesCouldTalk

City of Hamilton. (2026). *State of the urban forest report (2026)* (Appendix A to Report PW26014). https://www.hamilton.ca/sites/default/files/2026-02/urban-forest-strategy-state-forest-report-2026.pdf

CityLAB Hamilton. (2024, January 22). *Helping the tree canopy: Tree app pilot project*. https://www.citylabhamilton.com/winter-2024-projects/2024/1/22/tree-data-collection-and-tweeting-pilot-project

CityLAB Hamilton. (2026, May 11). *Open for matching: "If Trees Could Talk" data analysis*. https://www.citylabhamilton.com/active-challenges/iftreescouldtalkdata

Esperon-Rodriguez, M., Tjoelker, M. G., Lenoir, J., Baumgartner, J. B., Beaumont, L. J., Nipperess, D. A., Power, S. A., Richard, B., Rymer, P. D., & Gallagher, R. V. (2022). Climate change increases global risk to urban forests. *Nature Climate Change*, *12*(10), 950–955. https://doi.org/10.1038/s41558-022-01465-8

Harvard Forest. (n.d.). *What is a witness tree?* https://harvardforest.fas.harvard.edu/witness-tree-social-media-project